

5.2.56

EE25BTECH11054 - S.Harsha Vardhan

Question:

Solve the following system of linear equations.

$$\frac{5}{x-1} + \frac{1}{y-2} = 2 \quad \frac{6}{x-1} - \frac{3}{y-2} = 1$$

Solution:

Let

$$u = \frac{1}{x-1}, \quad v = \frac{1}{y-2}. \quad (1)$$

The given system becomes a linear system in terms of u and v :

$$5u + v = 2 \quad (2)$$

$$6u - 3v = 1 \quad (3)$$

In matrix form, this is $\mathbf{Ax} = \mathbf{b}$, where:

$$\begin{pmatrix} 5 & 1 \\ 6 & -3 \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}. \quad (4)$$

Let

$$\mathbf{A} = \begin{pmatrix} 5 & 1 \\ 6 & -3 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}. \quad (5)$$

We can solve this using Gauss-Jordan elimination on the augmented matrix $(\mathbf{A}|\mathbf{b})$.

$$\left(\begin{array}{cc|c} 5 & 1 & 2 \\ 6 & -3 & 1 \end{array} \right) \xrightarrow{R_1 \rightarrow \frac{1}{5}R_1} \left(\begin{array}{cc|c} 1 & \frac{1}{5} & \frac{2}{5} \\ 6 & -3 & 1 \end{array} \right) \xrightarrow{R_2 \rightarrow R_2 - 6R_1} \left(\begin{array}{cc|c} 1 & \frac{1}{5} & \frac{2}{5} \\ 0 & -\frac{21}{5} & -\frac{7}{5} \end{array} \right) \quad (6)$$

$$\left(\begin{array}{cc|c} 1 & \frac{1}{5} & \frac{2}{5} \\ 0 & -\frac{21}{5} & -\frac{7}{5} \end{array} \right) \xrightarrow{R_2 \rightarrow -\frac{5}{21}R_2} \left(\begin{array}{cc|c} 1 & \frac{1}{5} & \frac{2}{5} \\ 0 & 1 & \frac{1}{3} \end{array} \right) \xrightarrow{R_1 \rightarrow R_1 - \frac{1}{5}R_2} \left(\begin{array}{cc|c} 1 & 0 & \frac{1}{3} \\ 0 & 1 & \frac{1}{3} \end{array} \right) \quad (7)$$

From the reduced row echelon form, we have the solution for u and v :

$$\begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} \frac{1}{3} \\ \frac{1}{3} \end{pmatrix} \quad (8)$$

Now, we back-substitute to find x and y :

$$u = \frac{1}{x-1} = \frac{1}{3} \implies x-1 = 3 \implies x = 4, \quad (9)$$

$$v = \frac{1}{y-2} = \frac{1}{3} \implies y-2 = 3 \implies y = 5. \quad (10)$$

Thus, the final solution is:

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 4 \\ 5 \end{pmatrix}. \quad (11)$$

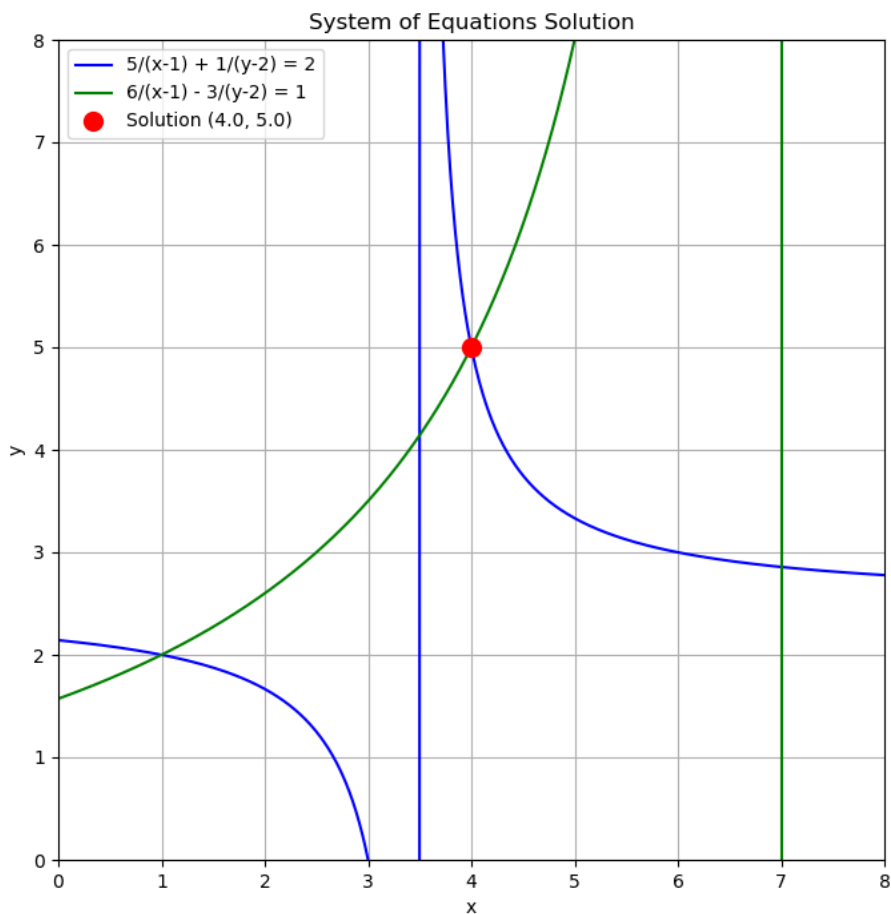


Fig. 0.1